Type MS · Factor theorem: show a factor or find a constant

Answer 1 of 10 · Question 3 part a · 2 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

3 (a)	$x = 4$ $64p - 496 + 100 + 12 = 0$ $64p = 384$ $p = 6$ *	M1 A1 (2)
(b)	$(x-4)(6x^2 - 7x - 3)$ $(x-4)(2x-3)(3x+1)$ $x = 4, \frac{3}{2}, -\frac{1}{3}$	M1 A1 M1 A1 (4)
		[6]

Part	Mark	Additional Guidance
(a)	M1	For substituting $x = 4$ into the **given** expression, equating the expression = 0 **and** attempting to solve for p
	A1	For $p = 6$ * This is a show question so every step must be seen
(b)	M1	For attempting to divide $6x^3 - 31x^2 + 25x + 12$ by $(x-4)$ $\Rightarrow x-4\overline{)6x^3 - 31x^2 + 25x + 12}$ $6x^2 - 7x + k$ (k is an integer)
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$ | Condone $\left(x - \frac{3}{2}\right)\left(x + \frac{1}{3}\right)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	ALT – equates coefficients	
	M1	For stating $6x^3 - 31x^2 + 25x + 12 = (x-4)(Ax^2 + Bx + C) \Rightarrow$ $6x^3 - 31x^2 + 25x + 12 = Ax^3 + x^2(B - 4A) + x(C - 4B) - 4C$ Minimum required is $A = 6,\ B = -7$ and $C = k$
	A1	For $A = 6,\ B = -7$ and $C = -3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	ALT – by inspection	
	M1	For finding the **quadratic** factor minimum required is $[(x-4)](6x^2 - 7x + k)$
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	Evidence of the 3TQ seen is required in part (b) $(x-4)(2x-3)(3x+1) = 0 \Rightarrow x = 4,\ \frac{3}{2},\ -\frac{1}{3}$ **is M0**	

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Question number	Scheme	Marks

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Question number	Scheme	Marks
3 (a)	$x = 4$ $64p - 496 + 100 + 12 = 0$ $64p = 384$ $p = 6$ *	 M1 A1 (2)
(b)	$(x-4)(6x^2 - 7x - 3)$ $(x-4)(2x-3)(3x+1)$ $x = 4, \frac{3}{2}, -\frac{1}{3}$	M1 A1 M1 A1 (4)
		[6]

Part	Mark	Additional Guidance
(a)	M1	For substituting $x = 4$ into the **given** expression, equating the expression = 0 **and** attempting to solve for p
	A1	For $p = 6$ * This is a show question so every step must be seen
(b)	M1	For attempting to divide $6x^3 - 31x^2 + 25x + 12$ by $(x-4)$ $\Rightarrow x-4\overline{)6x^3 - 31x^2 + 25x + 12}$ $6x^2 - 7x + k$ (k is an integer)
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$ | Condone $\left(x - \frac{3}{2}\right)\left(x + \frac{1}{3}\right)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	ALT – equates coefficients	
	M1	For stating $6x^3 - 31x^2 + 25x + 12 = (x-4)(Ax^2 + Bx + C) \Rightarrow$ $6x^3 - 31x^2 + 25x + 12 = Ax^3 + x^2(B - 4A) + x(C - 4B) - 4C$ Minimum required is $A = 6$, $B = -7$ and $C = k$
	A1	For $A = 6$, $B = -7$ and $C = -3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	ALT – by inspection	
	M1	For finding the **quadratic** factor minimum required is $[(x-4)](6x^2 - 7x + k)$
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	Evidence of the 3TQ seen is required in part (b)	
	$(x-4)(2x-3)(3x+1) = 0 \Rightarrow x = 4, \frac{3}{2}, -\frac{1}{3}$ **is M0**	

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Answer 1 of 10 · Question 3 part a · 2 marks

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Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 2 of 10 · Question 9 part b.i · 1 mark

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Number		
9(a)	$f'(x)=\lambda(x-2)(3x+1)$	M1
	$f(x)=\int\lambda(x-2)(3x+1)\,dx=\left[\int\lambda(3x^2-5x-2)\right]dx=\lambda\left(\frac{3x^3}{3}-\frac{5x^2}{2}-2x+k\right)$	M1A1
	At $(2,-9)$ $-9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=3$	M1
	$a=-5,b=-4,c=3\Rightarrow\{f(x)=2x^3-5x^2-4x+3\}$ *	A1*cso [5]
	ALT $f(x)=2x^3+ax^2+bx+c$ $f'(x)=6x^2+2ax+b$	[M1
	Both $\begin{cases}0=6(2^2)+2(2)a+b\\0=6\times\left(-\frac{1}{3}\right)^2+2\times\left(-\frac{1}{3}\right)\times a+b\end{cases}$	M1A1
	$\begin{cases}72+12a+3b=0 \text{ and } 12-12a+18b=0\Rightarrow a=...,b=...\\\text{At } (2,9)\quad -9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=...\end{cases}$	M1
	$a=-5,b=-4,c=3\Rightarrow\{f(x)=2x^3-5x^2-4x+3\}$ *	A1*cso]
(b)(i)	$f(-1)=2(-1)^3-5(-1)^2-4(-1)+3=0$ oe	B1 [1]
(b)(ii)	$x+1\overline{)2x^3-5x^2-4x+3}$ $\quad 2x^2-7x+3$	M1
	$2x^2-7x+3=(2x-1)(x-3)\Rightarrow(x+1)(x-3)(2x-1)$	M1A1 [3]
(c)	$A=\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$	M1
	$A=\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{-1}^{\frac{1}{2}}+\left\lvert\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{\frac{1}{2}}^{3}\right\rvert$	M1
	$A=\left\{\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]-\left[\frac{1}{2}.(-1)^4-\frac{5}{3}.(-1)^3-2.(-1)^2+3.(-1)\right]\right\}$ $+\left\lvert\left\{\left[\frac{1}{2}.(3)^4-\frac{5}{3}.(3)^3-2.(3)^2+3(3)\right]-\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]\right\}\right\rvert$	M1
	$A=\left\{\frac{117}{32}+\left\lvert-\frac{1375}{96}\right\rvert\right\}=\frac{863}{48}$	A1 [4]
		Total 13 marks

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Answer 2 of 10 · Question 9 part b.i · 1 mark

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Part	Mark	Additional Guidance
(a)	M1	For forming an expression for $f'(x)$ using the two turning points given. Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	Attempts to integrate, see General Guidance for the definition of an attempt Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	A1	Correct integration including +c, Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	$\lambda = 2$ is stated or implied, $\lambda k = c$ Substitution of $(2,-9)$ to their $f(x)$ (Does not have to be simplified) or substitution of $\left(-\frac{1}{3}, \frac{100}{27}\right)$ (Does not have to be simplified)
	A1*cso	Obtains the **given** answer with no errors in the working
	ALT - starts with $f(x) = 2x^3 + ax^2 + bx + c$	
	M1	Attempts to differentiate, see General Guidance for the definition of an attempt
	M1	Equates differentiated expression to 0 and substitutes $x = 2$ **and** equates differentiated expression to 0 and substitutes $x = -\frac{1}{3}$
	A1	Two correct equations, simplified or unsimplified
	M1	Attempts to solve two **correct** equations simultaneously (including using calculators) to find a, b **and** substitution of their $a, b, (2,-9)$ to $f(x)$ to find c or substitution of their $a, b, \left(-\frac{1}{3}, \frac{100}{27}\right)$ to $f(x)$ to find c
	A1*cso	Obtains the **given** answer with no errors in the working
(b)(i)	B1	Demonstrating that $x+1$ is a factor $f(-1) = 2(-1)^3 - 5(-1)^2 - 4(-1) + 3 = 0$ (factor theorem) If shown via polynomial division, the division must be fully correct and obtains 0 as the remainder.
(b)(ii)	M1	Dividing by $(x+1)$ to obtain a three-term quadratic quotient: $2x^2 - 7x + ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line : e.g. $(x+1)(x-3)(2x-1)$
	ALT I – equates coefficients	
	M1	For stating $2x^3 - 5x^2 - 4x + 3 = (x+1)(Ax^2 + Bx + C) \Rightarrow$ $2x^3 - 5x^2 - 4x + 3 = Ax^3 + (A+B)x^2 + (B+C)x + C \Rightarrow$ $A = 2, B = -7, C = ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$
	ALT II - inspection	
	M1	For finding the quadratic factor. Minimum required is $(x+1)(2x^2 - 7x + C)$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$

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(c)	M1	For identifying the correct limits, uses a correct strategy to find the area above the x-axis and below the x-axis: $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$ or $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx-\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx$ or implied by their working
	M1	For an attempt to integrate $2x^3-5x^2-4x+3$ See General Guidance for the definition of an attempt.
	M1	For clear substitution of **correct** limits into any **one** of their integrated expressions (must be a changed expression). Can be implied by **one** of the **correct exact** areas found from a **correctly** integrated expression
	A1	For correctly obtaining $\frac{863}{48}$ or $17\frac{47}{48}$

Type MS · Factor theorem: show a factor or find a constant

Answer 2 of 10 · Question 9 part b.i · 1 mark

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Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 3 of 10 · Question 1 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Question		Working	Marks
1	(a)	$-3\times q=-12\Rightarrow q=4$	M1A1 [2]
	(b)	$b^2-4ac<0,\ \ (p-2)^2-4\times1\times'4'<0$ To find critical values; $p^2-4p-12=0\Rightarrow(p-6)(p+2)=0\Rightarrow p=6,\ -2$ (Chooses inside region from their critical values) $-2<p<6$	M1 dM1A1 ddM1 A1 [5]
	Alternative		
	(b)	$b^2-4ac<0,\ \ (p-2)^2-4\times1\times'4'<0$ $(p-2)^2<16\Rightarrow-4<(p-2)<4$ $-2<p<6$	M1 dM1A1 ddM1A1
			Total 7 marks

(a)	M1	Substitute $x=0$ and $f(x)=-12$ and proceed to eliminate the term in p to obtain a linear equation in q.
	A1	$q=4$ (score M1 A1 if no working shown)
(b)	For the first 3 marks allow $b^2-4ac<0$ or $b^2-4ac=0$ or $b^2-4ac>0$	
	M1	Attempt to use b^2-4ac (<0) with $b=p-2, a=1$ and $c=4$ (ft from (a)). Correct formula shown and used or correct substitution.
	dM1	Attempt to solve their quadratic inequality or equation. Depends on previous M mark. See general guidance for solving quadratic equations.
	A1	Correct critical values, $(p=)-2, (p=)6$
	ddM1	Correct use of their critical values, showing that the inside region has been selected. Both M marks needed. Allow with $\leq$ or $<$ or set language statement.
	A1	$-2<p<6$ must be in terms of p. Allow $p\in(-2,6)$ or $-2<p$ AND p < 6 or $\{-2<p\ \cap\ p<6\}$
Alternative		
(b)	For the first 3 marks allow $b^2-4ac<0$ or $b^2-4ac=0$ or $b^2-4ac>0$	
	M1	Attempt to use $b^2-4ac<0$ with $b=p-2, a=1$ and $c=4$ (ft from (a)). Correct formula shown and used or correct substitution.
	dM1	Write in the form $(p-2)^2<16$ and attempt to take square root of both sides (both positive and negative options are needed), or as $(p-2)^2-16<0$ and attempt to factorise $[(p-2)-4][[(p-2)+4]]<0$ Depends on first M mark.
	A1	Correct critical values, $p=-2, p=6$ (Accept if wrong variable is used.)
	ddM1	Correct use of their critical values, showing that the inside region has been selected. Both M marks needed. Allow with $\leq$ or $<$ or set language statement.
	A1	$-2<p<6$ (in terms of p) with options as above.

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Answer 3 of 10 · Question 1 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 4 of 10 · Question 2 part b · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2 (a)	$f(1)=1+p+q=-12$	M1
	$p+q=-13$	A1
	$f(4)=64+4p+q=30$	M1
	$4p+q=-34$	A1
	$3p=-21$	M1
	$p=-7$ and $q=-6$	A1 (6)
(b)	$3^3-7\times3-6=27-21-6=0$ *	B1 cso (1)
(c)	$(x-3)(x^2+3x+2)$	M1
	$(x-3)(x+2)(x+1)$	M1 A1 (3)
(d)	$x=3$ $x=-2$ $x=-1$	B1 ft (1)
		[11]

	Notes
(a)	
M1	For substitution of 1 into f(x)
A1	For $p+q=-13$ oe
M1	For substitution of 4 into f(x)
A1	For $4p+q=-34$ oe
M1	For solving simultaneously
A1	$p=-7$ and $q=-6$
(b)	
B1cso	For substituting 3 into f(x) and obtaining the given result
(c)	
M1	For $(x-3)(x^2+3x+2)$
M1	For factorising the quadratic
A1	$(x-3)(x+2)(x+1)$
(d)	
B1ft	For $x=3$ $x=-2$ $x=-1$ or follow through part (c)

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Answer 4 of 10 · Question 2 part b · 1 mark

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Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 5 of 10 · Question 10 part a · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
10 (a)	$f(1) = 32(1^3) - 33(1) + 1 = 0$ *	B1 cso (1)
(b)	$(x-1)(32x^2+32x-1)=0$ A correct method shown to solve a quadratic	M1
	e.g. $\frac{-32 \pm \sqrt{32^2+4\times32}}{64}$	M1
	$(x=1)$ or $\frac{-4+3\sqrt{2}}{8}$ or $\frac{-4-3\sqrt{2}}{8}$ Accept decimals correct to 3 sf e.g. = 0.0303… or = –1.03 …	A1 A1 (4)
(c)	$\sqrt{x} = \frac{1}{8x}$	M1
	$p = \frac{1}{4}$	A1 (2)

	Notes
(a)	
B1 cso	For substitution of 1 into $f(x)$ to obtain the given result
(b)	
M1	For $(x-1)(32x^2+32x-1)=0$
M1	A correct method shown to solve a quadratic. If an algebraic method is not shown then M0A0A0 is awarded.
A1	For $\frac{-4+3\sqrt{2}}{8}$ Allow 0.0303 or better
A1	For $\frac{-4-3\sqrt{2}}{8}$ Allow -1.03 or better
(c)	
M1	For equating the two equations
A1	For $p = \frac{1}{4}$

Type MS · Factor theorem: show a factor or find a constant

Answer 5 of 10 · Question 10 part a · 1 mark

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(d)	$\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x = \pi\left[\frac{x^2}{2}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right)$	A1
	$\pi\int_{\frac{1}{4}}^{a} \frac{1}{64x^2}\,\mathrm{d}x = \pi\left[-\frac{1}{64x}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{1}{16} - \frac{1}{64a}\right)$	A1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right) - \pi\left(\frac{1}{16} - \frac{1}{64a}\right) = \frac{27\pi}{64}$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7)
	Alternative Method	
	$\pi\int_{\frac{1}{4}}^{a}\left(x - \frac{1}{64x^2}\right)\mathrm{d}x = \frac{27\pi}{64}$	M1
	$\int_{\frac{1}{4}}^{a}\left(64x - \frac{1}{x^2}\right)\mathrm{d}x = 27$	A1
	$\left[32x^2 + \frac{1}{x}\right]_{\frac{1}{4}}^{a} = 27$	M1 A1
	$\left(32a^2 + \frac{1}{a}\right) - (2+4) = 27$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7) **[14]**

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Answer 5 of 10 · Question 10 part a · 1 mark

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	Notes
(d)	
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x$ Ignore limits
A1	For $\pi\left(\frac{a^2}{2}-\frac{1}{32}\right)$
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a}\frac{1}{64x^2}\,\mathrm{d}x$ Ignore limits
A1	For $=\pi\left(\frac{1}{16}-\frac{1}{64a}\right)$
M1	Dependant on at least one previous M mark being awarded - for subtraction of the two integrals.
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$
	Alternative
M1	For $\pi\int_{\frac{1}{4}}^{a}\left(x-\frac{1}{64x^2}\right)\mathrm{d}x=\frac{27\pi}{64}$ Ignore limits
A1	For $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x=27$ Ignore limits
M1	For an attempt to integrate $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x$ Ignore limits
A1	$\left[32x^2+\frac{1}{x}\right]_{\frac{1}{4}}^{a}=27$
M1	Dependant on at least one previous M mark being awarded - for correct substitution of the limits
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$

Type MS · Factor theorem: show a factor or find a constant

Answer 6 of 10 · Question 6 part a · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question number		
6 (a)	$f(3) = 27 + 9p + 9 - 30 + q = 0$	M1 A1
	$9p + q + 6 = 0$ *	A1 cso
		(3)
(b)	$f(-p) = -p^3 + p^2(p+1) + 10p + q = 0$	M1 A1
	$p^2 + 10p + q = 0$ *	A1 cso
		(3)
(c)	$p^2 + 10p - 9p - 6 = 0$	M1
	$p^2 + p - 6 = 0$	A1
	$(p+3)(p-2) = 0$	M1
	$p = 2 \quad q = -24$	A1 A1
		(5)
(d)	$(x+a)(x-3)(x+2)$	
	So $-3 \times 2 \times a = -24 \quad a = 4$	M1
	$(x+4)(x-3)(x+2)$	A1
		(2)
		Total 13 marks

Part	Mark	Guidance
General guidance for marking parts (a) and (b) • For the award of full marks in parts (a) and/or (b) you **must** see = 0 used in a line of working **before** the final answer. • If a candidate does not use = 0 in either parts (a) or (b) [except in the final line – which is a given answer] deduct the M mark (and the subsequent A marks) in only the **first** occurrence of the absence.		
(a)	**M1**	For using $f(\pm 3) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $27 + 9p + 9 - 30 + q = 0$
	A1 cso	For obtaining the given equation $9p + q + 6 = 0$ * **Note: This is a show question. There must be no errors seen.**
(b)	**M1**	For use of $f(\pm p) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $-p^3 + p^2(p+1) + 10p + q = 0$

Type MS · Factor theorem: show a factor or find a constant

Answer 6 of 10 · Question 6 part a · 3 marks

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	A1 cso	For obtaining the correct given equation $p^2+10p+q=0$ * **Note: This is a show question. There must be no errors seen.**
(c)	**M1**	For attempting to solve the given two equations simultaneously to achieve a 3TQ in either p or q only. E.g. substitutes $q=\mp 9p\mp 6$ or $\left[p=\frac{\mp q\mp 6}{9} \text{ and } p^2=\frac{(\mp q\mp 6)^2}{81}\right]$ into $p^2+10p+q=0$ This mark may be implied by the correct 3TQ
	A1	For the correct 3TQ $p^2+p-6=0$ or $q^2+3q-504=0$
	M1	For an attempt to solve their 3TQ in either p or q using factorisation, use of the formula or completing the square. See general guidance for the definition of an attempt. For example: $(p+3)(p-2)=0\Rightarrow p=\ldots$ or $(q+24)(q-21)=0\Rightarrow q=\ldots,\ \ldots$ If a candidate uses their calculator to solve their 3TQ, the final values must be correct for the award of this mark unless a valid method is seen.
	A1	For either the correct value of p OR the correct value of q $p=2$ or $q=-24$ Condone the presence $p=-3$, and/or $q=21$
	A1	For both the correct value of p (= 2) AND the correct value of q (= –24) Must reject $p=-3$, and/or $q=21$ if seen.
(d)	**M1**	[f(x) = $(x+a)(x-3)(x+'2')$] For attempting to find the value of a $-3\times'2'\times a='-24'\Rightarrow a=\ldots$ OR For an attempt using division with their values of p and q $x^2-x-6\overline{)x^3+3x^2-10x-24}$ or $x-3\overline{)x^3+3x^2-10x-24}$ Allow a quotient of $x+b$ or x^2+6x+b where b is a constant.
	A1	For the correct factorised expression $(x+4)(x-3)(x+2)$ which must be written out in full on one line.

Type MS · Factor theorem: show a factor or find a constant

Answer 6 of 10 · Question 6 part a · 3 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 7 of 10 · Question 6 part b · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question number		
6 (a)	$f(3) = 27 + 9p + 9 - 30 + q = 0$	M1 A1
	$9p + q + 6 = 0$ *	A1 cso
		(3)
(b)	$f(-p) = -p^3 + p^2(p+1) + 10p + q = 0$	M1 A1
	$p^2 + 10p + q = 0$ *	A1 cso
		(3)
(c)	$p^2 + 10p - 9p - 6 = 0$	M1
	$p^2 + p - 6 = 0$	A1
	$(p+3)(p-2) = 0$	M1
	$p = 2 \quad q = -24$	A1 A1
		(5)
(d)	$(x+a)(x-3)(x+2)$	
	So $-3 \times 2 \times a = -24 \quad a = 4$	M1
	$(x+4)(x-3)(x+2)$	A1
		(2)
		Total 13 marks

Part	Mark	Guidance
General guidance for marking parts (a) and (b) • For the award of full marks in parts (a) and/or (b) you **must** see = 0 used in a line of working **before** the final answer. • If a candidate does not use = 0 in either parts (a) or (b) [except in the final line – which is a given answer] deduct the M mark (and the subsequent A marks) in only the **first** occurrence of the absence.		
(a)	**M1**	For using $f(\pm 3) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $27 + 9p + 9 - 30 + q = 0$
	A1 cso	For obtaining the given equation $9p + q + 6 = 0$ * **Note: This is a show question. There must be no errors seen.**
(b)	**M1**	For use of $f(\pm p) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $-p^3 + p^2(p+1) + 10p + q = 0$

Type MS · Factor theorem: show a factor or find a constant

Answer 7 of 10 · Question 6 part b · 3 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

	A1 cso	For obtaining the correct given equation $p^2+10p+q=0$ * **Note: This is a show question. There must be no errors seen.**
(c)	**M1**	For attempting to solve the given two equations simultaneously to achieve a 3TQ in either p or q only. E.g. substitutes $q=\mp 9p\mp 6$ or $\left[p=\frac{\mp q\mp 6}{9} \text{ and } p^2=\frac{(\mp q\mp 6)^2}{81}\right]$ into $p^2+10p+q=0$ This mark may be implied by the correct 3TQ
	A1	For the correct 3TQ $p^2+p-6=0$ or $q^2+3q-504=0$
	M1	For an attempt to solve their 3TQ in either p or q using factorisation, use of the formula or completing the square. See general guidance for the definition of an attempt. For example: $(p+3)(p-2)=0\Rightarrow p=\ldots$ or $(q+24)(q-21)=0\Rightarrow q=\ldots,\ \ldots$ If a candidate uses their calculator to solve their 3TQ, the final values must be correct for the award of this mark unless a valid method is seen.
	A1	For either the correct value of p OR the correct value of q $p=2$ or $q=-24$ Condone the presence $p=-3$, and/or $q=21$
	A1	For both the correct value of p (= 2) AND the correct value of q (= –24) Must reject $p=-3$, and/or $q=21$ if seen.
(d)	**M1**	[f(x) = $(x+a)(x-3)(x+'2')$] For attempting to find the value of a $-3\times'2'\times a='-24'\Rightarrow a=\ldots$ OR For an attempt using division with their values of p and q $x^2-x-6\overline{)x^3+3x^2-10x-24}$ or $x-3\overline{)x^3+3x^2-10x-24}$ Allow a quotient of $x+b$ or x^2+6x+b where b is a constant.
	A1	For the correct factorised expression $(x+4)(x-3)(x+2)$ which must be written out in full on one line.

Type MS · Factor theorem: show a factor or find a constant

Answer 7 of 10 · Question 6 part b · 3 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 8 of 10 · Question 1 part a · 2 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

1. Factorisation:

$(x^2 + bx + c) = (x + p)(x + q)$,where $|pq| = |c|$ leading to $x =$….

$(ax^2 + bx + c) = (mx + p)(nx + q)$ where$|pq| = |c|$ and $|mn| = |a|$leading to $x =$….

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x =$….

3. Completing the square:

$x^2 + bx + c = 0$: $\left(x \pm \frac{b}{2}\right)^2 \pm q \pm c = 0, \quad q \neq 0$ leading to $x =$….

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $(x^n \rightarrow x^{n-1})$

2. Integration:

Power of at least one term increased by 1. $(x^n \rightarrow x^{n+1})$

Use of a formula:
Generally, the method mark is gained by **either** quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values **or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:
The rubric states "Without sufficient working, correct answers may be awarded no marks". General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

Type MS · Factor theorem: show a factor or find a constant

Answer 8 of 10 · Question 1 part a · 2 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks
1(a)	$f\left(\frac{2}{3}\right)=6\times\left(\frac{2}{3}\right)^3-13\times\left(\frac{2}{3}\right)^2+a\times\left(\frac{2}{3}\right)-10=0\Rightarrow\frac{2a}{3}=14$	M1
	$\Rightarrow a=21$*	A1 cso
	Alt – Polynomial division	[2]
	$3x-2\overline{)6x^3-13x^2+ax-10}\Rightarrow a-6=15\Rightarrow a=21$	[M1A1]
(b)	**Polynomial division**	
	$3x-2\overline{)6x^3-13x^2+21x-10}$	M1A1
	Equates coefficients	
	$(3x-2)(ax^2+bx+c)=6x^3-13x^2+21x-10$	
	$\Rightarrow 3a=6\Rightarrow a=2$	
	$\Rightarrow 3b-2a=-13\Rightarrow b=-3$	[M1
	$\Rightarrow -2c=-10\Rightarrow c=5$	
	Quadratic factor is $2x^2-3x+5$	A1]
	Conclusion	M1
	$b^2-4ac=(-3)^2-4\times2\times5=-31$	
	The discriminant is negative hence the quadratic has no real solutions and the only intersection with the x-axis is at $x=\frac{2}{3}$	A1 [4]
	ALT	
	Completes square:	
	$2x^2-3x+5=0\Rightarrow\left(x-\frac{3}{4}\right)^2=-\frac{31}{16}$	[M1
	$\left[\Rightarrow x=\frac{3}{4}\pm\sqrt{-\frac{31}{16}}\right]$	
	$-\frac{31}{16}<0$ so the quadratic has no real roots	
	The only intersection [root] with the x-axis is at $x=\frac{2}{3}$	A1]
		Total 6 marks

Type MS · Factor theorem: show a factor or find a constant

Answer 9 of 10 · Question 7 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1 \overline{)64x^3 - 64x^2 \{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$ $x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$ $x = \frac{1}{4}$ oe $x = \frac{3 \pm \sqrt{21}}{8}$ oe	M1 dM1 B1 A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe $(192 - 192r)r^2 = 9$ oe $192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	B1 M1 A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$ $a = 144$	M1 A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$ $1 - 0.25^n > \frac{1919}{1920}$ $0.25^n < 0.00052083...$ $n > \frac{\log 0.00052083}{\log 0.25}$ $n > 5.45... \Rightarrow n = 6$	M1 dM1 M1 A1 [4]
		Total 16 marks

Type MS · Factor theorem: show a factor or find a constant

Answer 9 of 10 · Question 7 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find *C*. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find *x*, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Factor theorem: show a factor or find a constant

Answer 9 of 10 · Question 7 part a · 2 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factor theorem: show a factor or find a constant

Answer 10 of 10 · Question 10 part a · 2 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

10 (a)	$f\left(\frac{3}{4}\right)=16\left(\frac{3}{4}\right)^3+11\left(\frac{3}{4}\right)-15$ * $\left(\frac{27}{4}+\frac{33}{4}-15\right)=0$	M1 A1cso [2]
(b)(i)	$[\sin 2\theta=\sin(\theta+\theta)]=\sin\theta\cos\theta+\sin\theta\cos\theta$	M1
	$(\sin 2\theta=)2\sin\theta\cos\theta$ *	A1cso [2]
(b)(ii)	$[\cos 2\theta=\cos(\theta+\theta)=]\cos\theta\cos\theta-\sin\theta\sin\theta=\cos^2\theta-\sin^2\theta$	M1
	$=\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$ *	M1 A1cso [3]
(c)	$27\cos\theta(2\cos^2\theta-1)+19\sin\theta(2\sin\theta\cos\theta)-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38(1-\cos^2\theta)\cos\theta-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38\cos\theta-38\cos^3\theta-15(=0)$ $(16\cos^3\theta+11\cos\theta-15=0)$ oe	A1
	(When $x=\cos\theta$) $16x^3+11x-15=0$ *	A1cso [4]
(d)	$(16x^3+11x-15=0\Rightarrow(4x-3)(4x^2+3x+5)=0)\rightarrow 4x^2+3x+5$	M1
	$b^2-4ac=9-80<0$ (So no real roots)	M1 A1
	(Only solution is) $4x-3=0$ or $4\cos\theta-3=0$ so $\cos\theta=\frac{3}{4}$ *	A1cso [4]
		Total 15 marks

Type MS · Factor theorem: show a factor or find a constant

Answer 10 of 10 · Question 10 part a · 2 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks